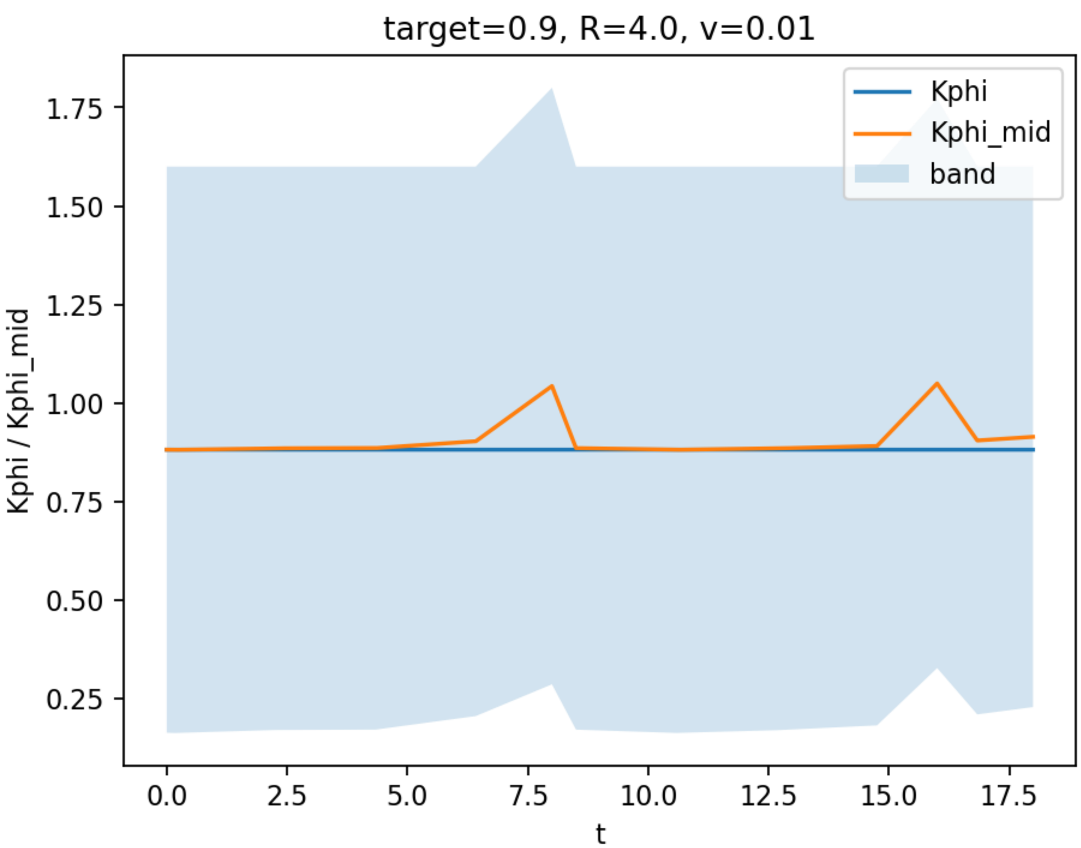
Phase-25 — Controller policy (rate-limit + hysteresis) & time-to-slip

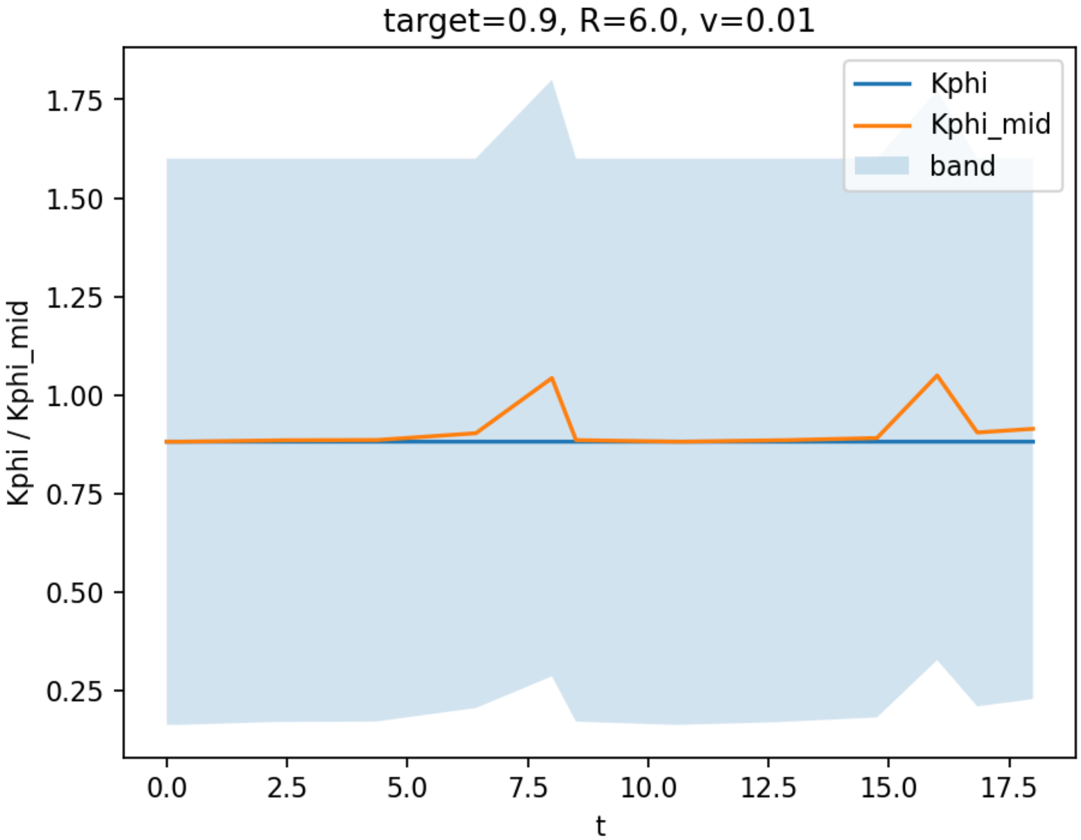
- Summary: `outputs/phase25/p25_summary.csv`
  ## Method We track $K_\phi(t)$ toward the band midpoint $K_{\text{mid}}(\omega^2)$ from Phase-23. The allowable error uses Phase-22 band width: half-width $= \frac{1}{2}$ band_width. A deadband $h = f_h \cdot \frac{1}{2}$ band_width disables actuation for small errors. The actuator is rate-limited by $R = \max |dK_\phi/dt|$. We declare slip when $|e(t)| = |K_{\text{mid}} - K_\phi| \geq f_s \cdot \frac{1}{2}$ band_width.

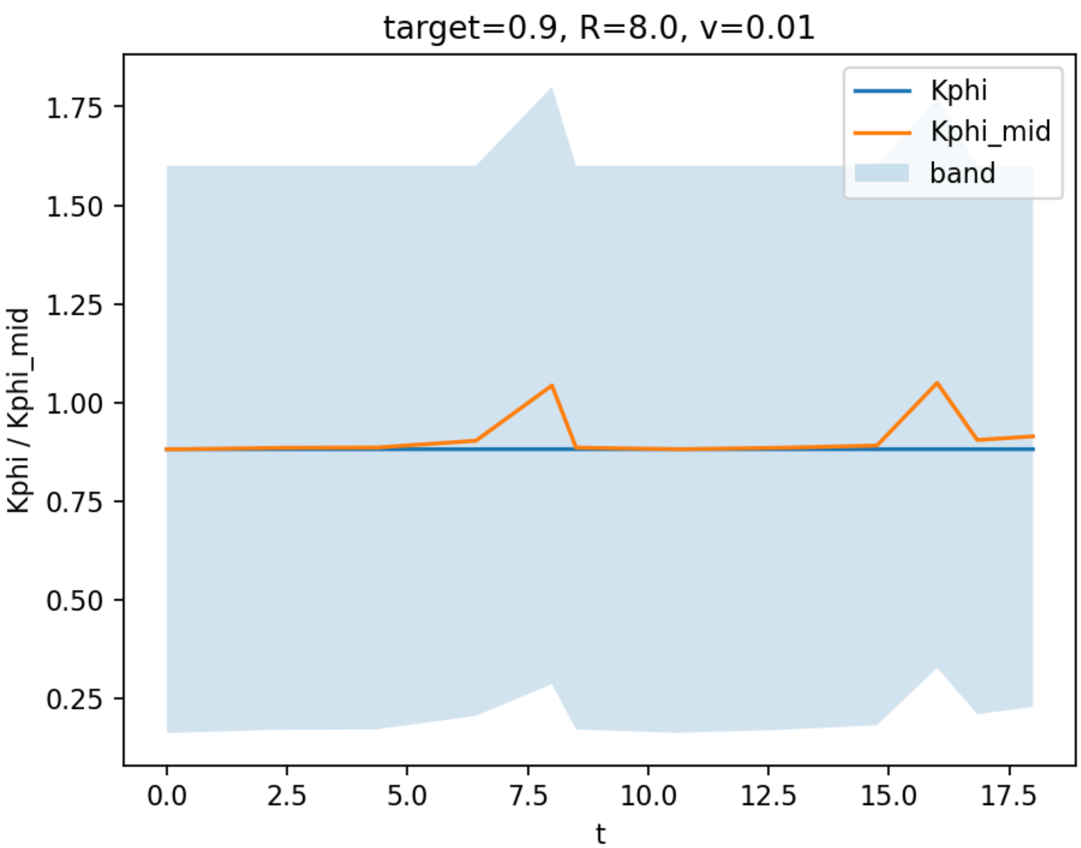
Runs

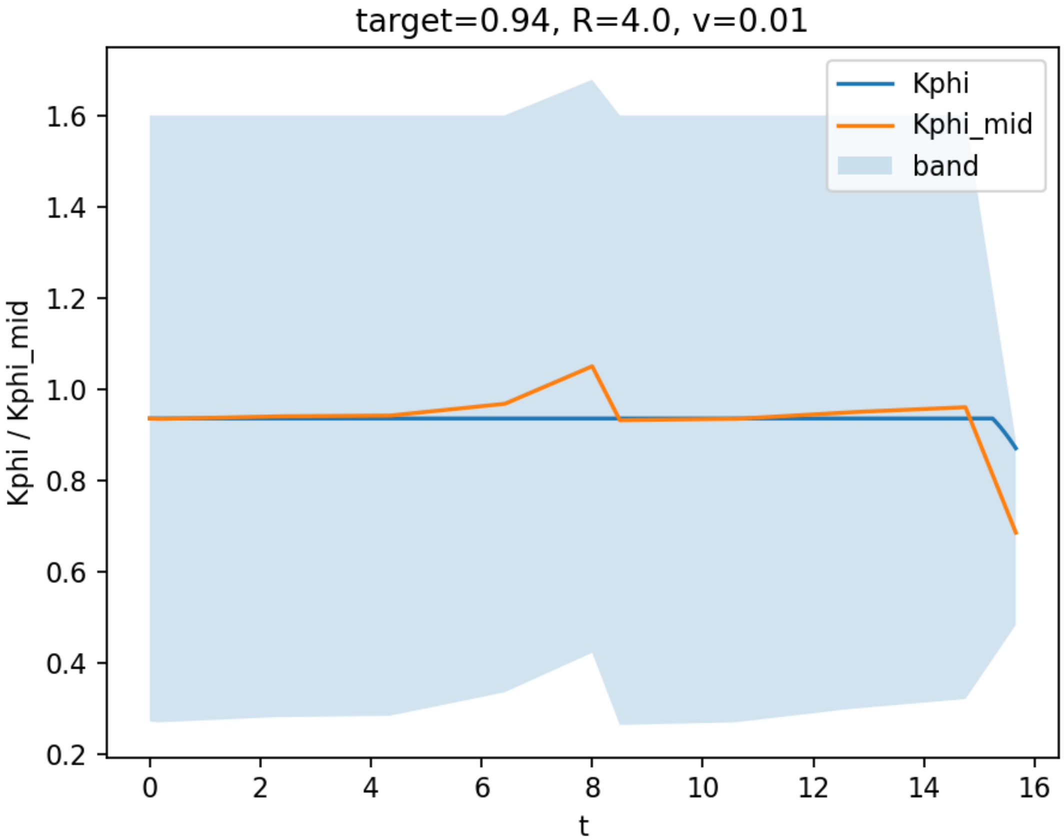
target	R	v	w2_start	w2_end	slipped	t_slip	w2_slip
0.90	4	0.01	2.74	2.92	False		
0.90	6	0.01	2.74	2.92	False		
0.90	8	0.01	2.74	2.92	False		
0.94	4	0.01	2.74	2.92	True		
0.94	6	0.01	2.74	2.92	True		
0.94	8	0.01	2.74	2.92	True		
0.96	4	0.01	2.74	2.91	True		
0.96	6	0.01	2.74	2.91	True		
0.96	8	0.01	2.74	2.91	True		
0.98	4	0.01	2.74	2.87	True		
0.98	6	0.01	2.74	2.87	True		
0.98	8	0.01	2.74	2.87	True		

Example time series

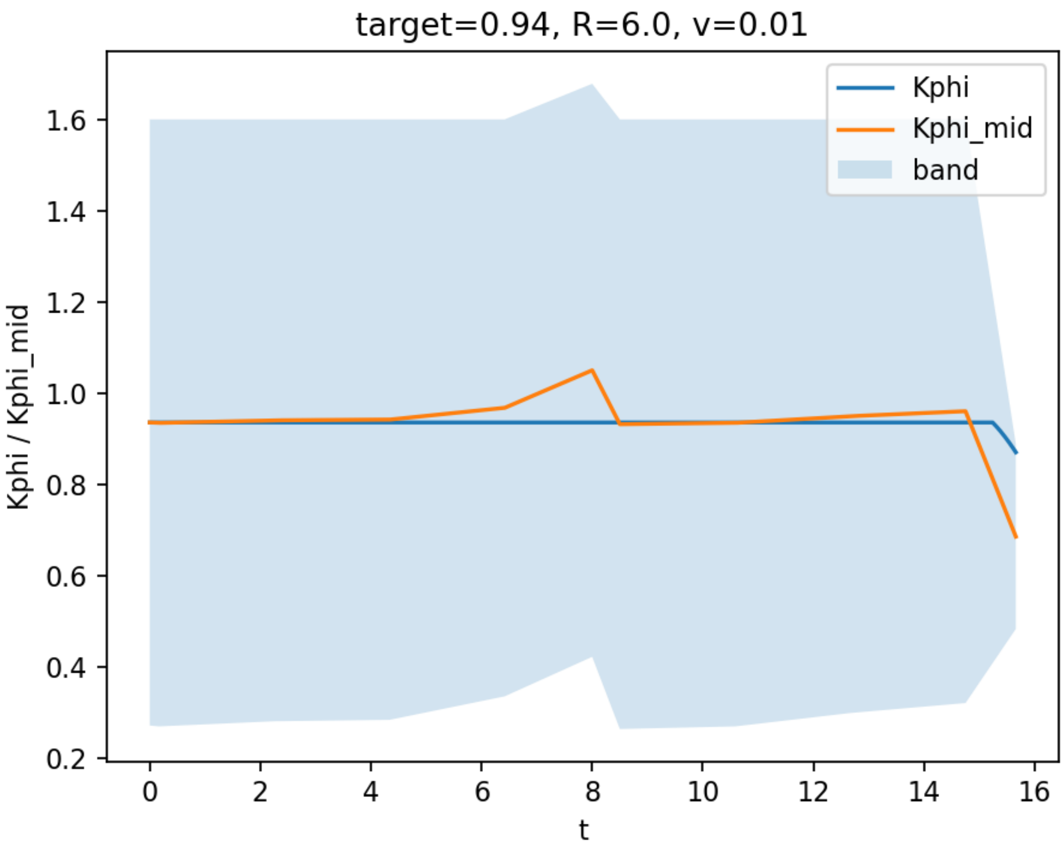



target=0.9, R=6.0, v=0.01
Kphi
Kphi_mid
band
Kphi / Kphi_mid
t
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

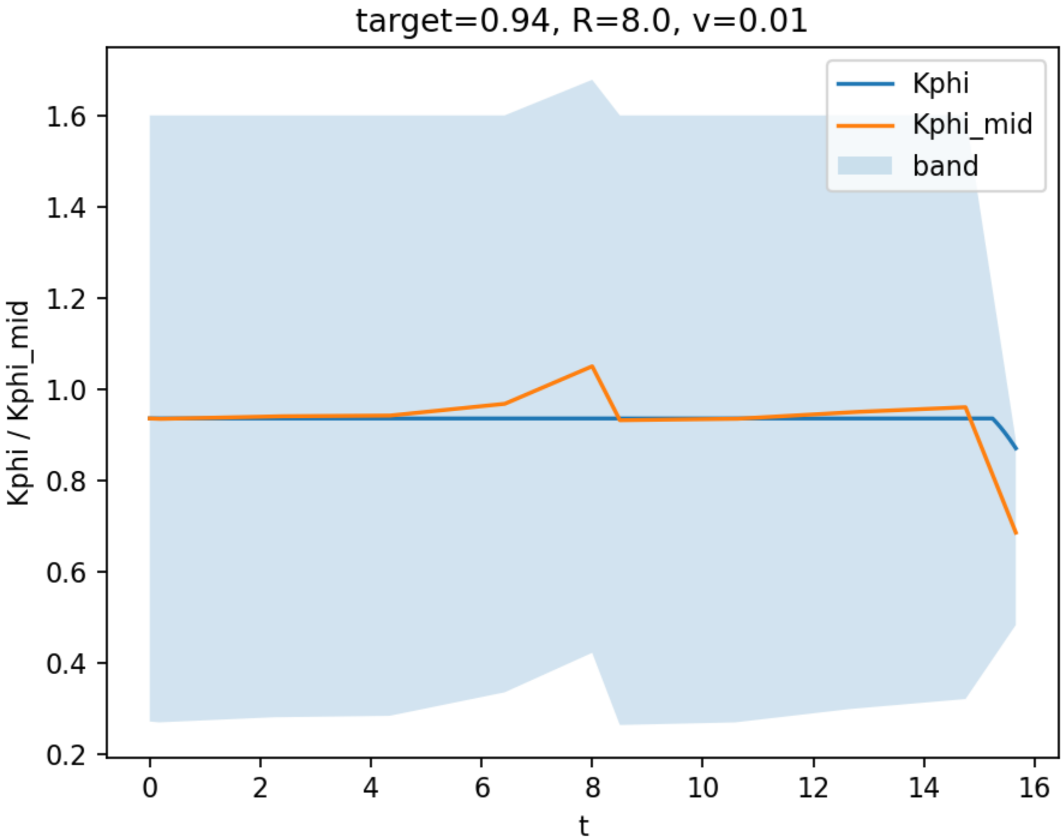

target=0.9, R=8.0, v=0.01
Kphi
Kphi_mid
band
Kphi / Kphi_mid
t
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5



target=0.94, R=4.0, v=0.01
Kphi
Kphi_mid
band
Kphi / Kphi_mid
t



target=0.94, R=6.0, v=0.01
Kphi
Kphi_mid
band
Kphi / Kphi_mid
t


target=0.94, R=8.0, v=0.01
Kphi
Kphi_mid
band
Kphi / Kphi_mid
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
0
2
4
6
8
10
12
14
16
t